SYSE-822 Applied Aerodynamics

Project-I Due: Oct 26th, 2026

Develop a custom numerical solver in Python / MATLAB to simulate inviscid, incompressible potential flow over arbitrary 2D airfoil geometries (such as NACA 4-digit series) using the Panel Method.

Project Evaluation Rubric & Deliverables

The project will be evaluated based on following primary components:

1. 10 Minutes presentation 15%
2. Code understanding 30%

3. Pressure Coefficient (C_p) Graph 15%
4. Lift Coefficient (C_L) Evaluation (Different α) 15%
5. Drag Coefficient (C_D) Analysis 05%
6. Velocity & Pressure Field Visualizations 20%